

Treasuries (which are “off the run”).²³ The spread between these two types of bonds varies over time reflecting liquidity conditions in Treasury markets.²⁴ (For more details, see chapter 9.)

There were pronounced illiquidity effects in Treasuries during the 2008 to 2009 financial crisis. Treasury bonds and notes are identical, except that the U.S. Treasury issues bonds with original maturities of twenty to thirty years and notes originally carry maturities of one to ten years. But after ten years, a Treasury bond originally carrying a twenty-year maturity is the same as a Treasury note. If the maturities are the same, whether this particular security is bond or a note should make no difference. During the financial crisis Treasury bond prices with the same maturity as Treasury notes had prices that were more than 5% lower—these are large illiquidity effects in one of the world’s most important and liquid markets.²⁵

Corporate Bonds

Corporate bonds that trade less frequently or have larger bid–ask spreads have higher returns. Chen, Lesmond, and Wei (2007) find that illiquidity risk explains 7% of the variation across yields of investment-grade bonds. Illiquidity accounts for 22% of the variation in junk bond yields; for these bonds, a one basis point rise in bid–ask spreads increases yield spreads by more than two basis points.²⁶

Equities

A large literature finds that many illiquidity variables predict returns in equity markets, with less liquid stocks having higher returns.²⁷ These variables include bid–ask spreads, volume, volume signed by whether trades are buyer or seller initiated, turnover, the ratio of absolute returns to dollar volume (commonly called the “Amihud measure” based on his paper of 2002), the price impact of large trades, informed trading measures (which gauge adverse, informed trading; see below), quote size and depth, the frequency of trades, how often there are “zero” returns (in more liquid markets returns will bounce up and down), and return autocorrelations (which are a measure of stale prices).²⁸ These are all illiquidity characteristics, which are properties unique to an individual stock. There are

²³The on-the-run bonds are more expensive because they can be used as collateral for borrowing funds in the repo market. This is called “*specialness*.” See Duffie (1996).

²⁴See Goyenko, Subrahmanyam, and Ukhov (2011).

²⁵See Musto, Nini, and Schwarz (2011).

²⁶See also chapter 9, Bao, Pan, and Wang (2011), Lin, Wang, and Wu (2011), and Dick-Nielsen, Feldhutter, and Lando (2012).

²⁷See the summary article by Amihud, Mendelson, and Pedersen (2005).

²⁸Sorting stocks on all these variables results in spreads in average returns. But some of these illiquidity measures produce spreads in expected returns opposite to an illiquidity risk premium. Stocks with higher than average (normalized) volume, for example, tend to have lower future returns as shown by Gervais, Kaniel, and Mingelgrin (2001).